

Julián Sierra Salamanca

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in [Julián Sierra](#)

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PROFESSIONAL PROFILE

Physical engineer currently pursuing a master's degree in applied mathematics. Experienced in mathematical modeling for process characterization and optimization, as well as machine learning models for prediction and customer segmentation. Interested in physical and econophysical modeling, computational simulation, and machine learning techniques applied to engineering and R&D projects. Skilled at transforming complex problems into functional quantitative models by integrating theoretical and applied approaches. Seeking to contribute to the development of technologies and systems in which mathematical modeling and data analysis drive innovation.

EDUCATION

Feb 2025 — Present	M.Sc. Student in Applied Mathematics <i>National University of Colombia</i>	Bogotá
Jun 2015 — Nov 2022	B.Sc. in Physical Engineering <i>National University of Colombia</i>	Bogotá

PROFESSIONAL EXPERIENCE

Dec 2025 — Present Bogotá, D.C.	Applied Mathematician, Sierra Analytics • Conduct data mining and exploratory data analysis (EDA). • Develop econometric and dynamic models and simulations to support decision-making. • Optimize inventory management. • Train machine learning models for prediction, recommendation, and segmentation of variables of interest.
Sep 2023 — Mar 2024 Bogotá, D.C.	Junior Engineer — Hydraulic Design Engineer, Cesva S.A.S. • Performed hydraulic calculations and modeled flow systems for infrastructure projects. • Developed and optimized Python and MATLAB algorithms for data analysis. • Contributed to technical reports and presented results to clients. • Participated in technical project management and advised multidisciplinary teams. • Applied quantitative analysis and statistical modeling techniques to develop innovative solutions.
Sep 2022 — Apr 2023 Bogotá, D.C.	Technical Sales Consultant, Techno Skills Engineering Services S.A.S. • Provided technical consulting to clients, aligning solutions with project requirements. • Developed analytical models in Python and MATLAB to support data-driven decision-making. • Led project teams in implementing engineering solutions and ensuring timely delivery. • Performed statistical analysis and data visualization, improving client presentations. • Collaborated with cross-functional teams to improve project outcomes.

Jul 2021 — Sep 2025 Bogotá, D.C.	Mathematics and Physics Instructor, University and ICFES Exam Preparation — Ingenious S.A.S., Deming S.A.S., and Multipruebas S.A.S. - Designed and delivered engaging mathematics and physics lessons tailored to students' needs. - Conducted assessments to measure student performance and provide personalized feedback. - Collaborated with other educators to improve curricula and teaching strategies. - Implemented innovative teaching methods to facilitate understanding of complex concepts. - Facilitated workshops for parents and students to strengthen academic support.
Apr 2024 — Nov 2024 Bogotá, D.C.	Mathematics and Physics Teacher, Limca S.A.S. - Implemented active learning strategies that improved participation and comprehension in multilevel classrooms. - Developed students' logical reasoning and problem-solving skills, achieving measurable improvements in academic performance. - Led student groups, fostering a positive school environment and providing comprehensive socioemotional support.

RESEARCH EXPERIENCE

Sep 2023 — Feb 2024 Manizales, Caldas	Academic Researcher, Plasma Torch Project — MinCiencias Conducted a systematic review of scientific and technical literature, identifying key trends, relevant technologies, and knowledge gaps applicable to the project. Performed a rigorous technical feasibility analysis for implementing a combustion torch, defining implementation requirements, potential risks, and optimal operating conditions for its effective deployment.
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TECHNICAL SKILLS

Programming	Python, Julia, R, MATLAB, SQL
Visualization	Power BI, Matplotlib, Seaborn
Analysis	Statistical Modeling, Machine Learning, Quantitative Analysis
Simulation	EPANET, TensorFlow
Management	Technical Project Management, Applied Research

CERTIFICATIONS AND COURSES

In Progress	Specialized Program: Data Science <i>Coursera — University of the Andes</i> Online Comprehensive training in the ASUM-DM methodology for end-to-end data project management. The program covers descriptive, univariate, and bivariate statistical techniques for data analysis, hypothesis testing, and business problem-solving. It includes complete data preparation, exploration, transformation, and integration processes with quality management. The program culminates in the development and implementation of predictive machine learning models in Python, providing a practical approach from preprocessing through model evaluation. Introduction to Applied Data Science
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May 2023

Machine Learning Specialization

Coursera — DeepLearning.AI

Online

Certification in building machine learning models with Python, covering supervised learning (linear regression, logistic regression, decision trees, and random forests) and unsupervised learning (clustering and anomaly detection), culminating in the implementation of neural networks using TensorFlow.

- Verified Certificate

LANGUAGES

Spanish

Native

English

Intermediate (B1–B2)